

JESS Thermodynamic Database v8.9

Rhenium

24-Jan-23

Reaction No. 4864							
$\text{Re}+3 + 3\text{e-1} = \text{Re(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Inf. Dilution	E:0.3(0.1SD)	3	EES	775
2	25	0	Inf. Dilution	E:0.300(3SF)	5	CRV	6330

Reaction No. 5226							
$\text{Re(s)} + \text{e-1} = \text{Re-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Unknown	lgK:-2.4(2SF)	4	ENS	773

Reaction No. 5227							
$\text{Re+1} + 2\text{e-1} = \text{Re-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Unknown	lgK:-7.8(2SF)	4	ENS	773

Reaction No. 32876							
$\text{ReO4-2} + 2\text{H2O} + 2\text{e-1} = \text{ReO2(s)} + 4\text{OH-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-0.5(1SF)	3	CRV	7761

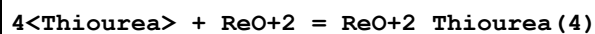
Reaction No. 36010							
$\text{Thiourea} + \text{ReO+2} = \text{ReO+2_Thiourea}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	4	HCl	lgK:2.4(0.05SD)	4	MIX	4140

Reaction No. 36011							
$2\text{Thiourea} + \text{ReO+2} = \text{ReO+2_Thiourea(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	4	HCl	lgK:5.5(0.2SD)	4	MIX	4140

Reaction No. 36012							
$3\text{Thiourea} + \text{ReO+2} = \text{ReO+2_Thiourea(3)}$							

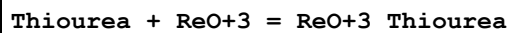
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	4	HCl	lgK:6.9(0.1SD)	4	MIX	4140

Reaction No. 36013



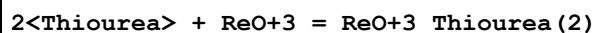
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	4	HCl	lgK:8.8(0.2SD)	4	MIX	4140

Reaction No. 36014



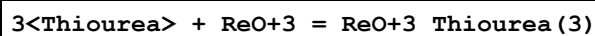
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	6	HCl	lgK:3.5(0.03SD)	4	MPS	4142

Reaction No. 36015



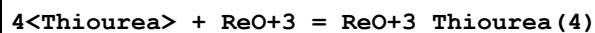
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	6	HCl	lgK:5.6(0.06SD)	4	MPS	4142

Reaction No. 36016



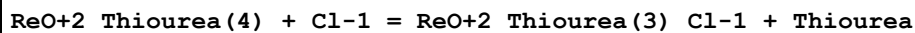
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	6	HCl	lgK:6.5(0.09SD)	4	MPS	4142

Reaction No. 36017



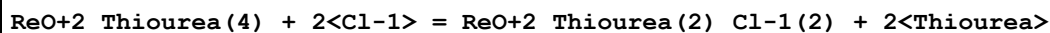
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	6	HCl	lgK:7.5(0.1SD)	4	MPS	4142

Reaction No. 42817



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	4	HCl	lgR:2.4(0.05SD)	4	MIX	4140

Reaction No. 42818



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	4	HCl	lgR:5.5(0.2SD)	4	MIX	4140

Reaction No. 42819							
$\text{ReO}+2_{\text{Thiourea}}(4) + 3\text{Cl-1} = \text{ReO}+2_{\text{Thiourea}}\text{Cl-1}(3) + 3\text{Thiourea}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	4	HCl	lgR:6.9(0.08SD)	4	MIX	4140

Reaction No. 42820							
$\text{ReO}+2_{\text{Thiourea}}(4) + 4\text{Cl-1} = \text{ReO}+2_{\text{Cl-1}}(4) + 4\text{Thiourea}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	4	HCl	lgR:8.8(0.2SD)	4	MIX	4140

Reaction No. 52478							
$\text{Re}+3(2)_{\text{Cl-1}}(2)_{\text{Propanoic-1}}(4) + \text{Br-1} = \text{Re}+3(2)_{\text{Br-1}}\text{Cl-1}_{\text{Propanoic-1}}(4) + \text{Cl-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25		MeCN	lgK:1.959(4SF)	4	MSP	4565

Reaction No. 52479							
$\text{Re}+3(2)_{\text{Br-1}}\text{Cl-1}_{\text{Propanoic-1}}(4) + \text{Br-1} = \text{Re}+3(2)_{\text{Br-1}}(2)_{\text{Propanoic-1}}(4) + \text{Cl-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25		MeCN	lgK:2.398(4SF)	4	MSP	4565

Reaction No. 54199							
$\text{Cl-1} + \text{Re}+2 = \text{Re}+2_{\text{Cl-1}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.32	H2SO4	lgK:2.0(2SF)	5	MKN	931

Reaction No. 54294							
$\text{Re}+3_{\text{Br-1}}(5) + \text{Br-1} = \text{Re}+3_{\text{Br-1}}(6)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15	3	HClO4	lgK:5.26(3SF)	4	MAG	931

Reaction No. 54667							
$\text{ReO}4-1 + 2\text{H}+1 + \text{e-1} = \text{ReO}3(\text{s}) + \text{H}2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.77(0.01SD)	2	MEF	775
2	25	0	Inf. Dilution	E:0.768(3SF)	2	CRV	6330

Reaction No. 54668							
$\text{ReO}4-1 + 4\text{H}+1 + 3\text{e-1} = \text{ReO}2(\text{s}) + 2\text{H}2\text{O}$							

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:29.4(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:0.51(0.01SD)	0	MEF	775
3	25	0	Inf. Dilution	E:0.510(3SF)	0	CRV	6330

Reaction No. 54669							
ReO4-1 + 8<H+1> + 7<e-1> = Re(s) + 4<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Inf. Dilution	E:0.37(0.01SD)	4	MDG	775
2	25	0	Inf. Dilution	E:0.368(3SF)	5	CRV	6330

Reaction No. 54670							
ReO3(s) + 2<H+1> + 2<e-1> = ReO2(s) + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.3(3SF)	1	EAE	
2	20	0	Inf. Dilution	E:0.4(0.1SD)	0	MDG	775

Reaction No. 54671							
ReO2(s) + 4<H+1> + 4<e-1> = Re(s) + 2<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.6(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:0.26(0.01SD)	0	MDG	775
3	25	0	Inf. Dilution	E:0.2513(4SF)	0	CRV	6330

Reaction No. 65649							
ReO4-1 + K+1 = K+1_ReO4-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.72(2SF)	4	CRV	552
2	25	0	Inf. Dilution	lgK:0.72(2SF)	4	MCN	7981

Reaction No. 67213							
7<As(s)> + 3<Re(s)> = Re3As7(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.53(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:15.42(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:11.27(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:8.778(4SF)	1	ENS	6422

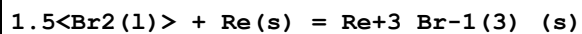
5	327	0	Inf. Dilution	lgK:7.116(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-21.18(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-22.80(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-22.81(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-22.81(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-22.81(4SF)	0	MTD	6422

Reaction No. 67214



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:118.9(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:118.0(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:84.47(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:64.39(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:51.05(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-162.1(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-184.4(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-184.0(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-183.4(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-182.7(4SF)	0	MTD	6422

Reaction No. 67215



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:27.44(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:27.25(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:18.41(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:12.70(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:8.957(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-37.44(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-42.00(4SF)	4	MTD	6422
8	27	0	Inf. Dilution	dH:-42.01(4SF)	4	MTD	6422
9	127	0	Inf. Dilution	dH:-52.50(4SF)	4	MTD	6422
10	227	0	Inf. Dilution	dH:-51.79(4SF)	4	MTD	6422
11	327	0	Inf. Dilution	dH:-50.95(4SF)	4	MTD	6422

Reaction No. 67216

1.5<Cl ₂ (g)> + Re(s) = Re+3_Cl-1(3)_ (s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:33.33(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:33.04(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:21.59(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:14.79(4SF)	3	ENS	6422
5	327	0	Inf. Dilution	lgK:10.31(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-188.0(4SF)kJ	3	CRV	6330
7	25	0	Inf. Dilution	dG:-45.47(4SF)	4	EFE	6422
8	25	0	Inf. Dilution	dH:-264.0(4SF)kJ	3	CRV	6330
9	25	0	Inf. Dilution	dH:-63.10(4SF)	4	MTD	6422
10	127	0	Inf. Dilution	dH:-62.57(4SF)	4	MTD	6422
11	227	0	Inf. Dilution	dH:-61.88(4SF)	3	MTD	6422
12	327	0	Inf. Dilution	dH:-61.09(4SF)	0	MTD	6422

Reaction No. 67217							
O ₂ (g) + Re(s) = ReO ₂ (s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:68.53(4SF)	3	ENS	6422
2	27	0	Inf. Dilution	lgK:68.04(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:48.51(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:36.80(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:29.02(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-93.49(4SF)	3	EFE	6422
7	25	0	Inf. Dilution	dG:-93.494(5SF)	3	CRV	31291
8	25	0	Inf. Dilution	dH:-107.3(4SF)	1	MTD	6422
9	127	0	Inf. Dilution	dH:-107.2(4SF)	0	MTD	6422
10	227	0	Inf. Dilution	dH:-107.0(4SF)	0	MTD	6422
11	327	0	Inf. Dilution	dH:-106.7(4SF)	0	MTD	6422

Reaction No. 67218							
1.5<O ₂ (g)> + Re(s) = ReO ₃ (s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:93.8(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:88.84(4SF)	0	ENS	6422
3	27	0	Inf. Dilution	lgK:88.21(4SF)	0	ENS	6422

4	127	0	Inf. Dilution	lgK:62.58 (4SF)	0	ENS	6422
5	227	0	Inf. Dilution	lgK:47.24 (4SF)	0	ENS	6422
6	327	0	Inf. Dilution	lgK:37.05 (4SF)	0	ENS	6422
7	25	0	Inf. Dilution	dG:-121.2 (4SF)	0	EFE	6422
8	25	0	Inf. Dilution	dG:-121.22 (5SF)	0	CRV	31291
9	25	0	Inf. Dilution	dH:-140.8 (4SF)	0	MTD	6422
10	127	0	Inf. Dilution	dH:-140.6 (4SF)	0	MTD	6422
11	227	0	Inf. Dilution	dH:-140.2 (4SF)	0	MTD	6422
12	327	0	Inf. Dilution	dH:-139.7 (4SF)	0	MTD	6422

Reaction No. 67219							
3.5<O2(g)> + 2<Re(s)> = Re2O7(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:190.8 (4SF)	3	ENS	6422
2	27	0	Inf. Dilution	lgK:189.4 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:134.5 (4SF)	2	ENS	6422
4	140	0	Inf. Dilution	lgK:129.3 (4SF)	2	ENS	6422
5	25	0	Inf. Dilution	dG:-260.3 (4SF)	3	EFE	6422
6	25	0	Inf. Dilution	dG:-260.33 (5SF)	3	CRV	31291
7	25	0	Inf. Dilution	dH:-301.9 (4SF)	2	MTD	6422
8	127	0	Inf. Dilution	dH:-301.4 (4SF)	1	MTD	6422
9	140	0	Inf. Dilution	dH:-301.3 (4SF)	1	MTD	6422

Reaction No. 67220							
Re(s) + 2<S(s)> = ReS2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:29.21 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:29.02 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:21.21 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:16.38 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:13.11 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-39.85 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-42.70 (4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-43.86 (4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-44.62 (4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-45.18 (4SF)	4	MTD	6422

Reaction No. 67221							
$2\text{Re(s)} + 7\text{S(s)} = \text{Re}_2\text{S}_7\text{(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:72.30 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:71.81 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:52.05 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:39.70 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:31.26 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-98.63 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-107.9 (4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-111.9 (4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-114.7 (4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-116.9 (4SF)	4	MTD	6422

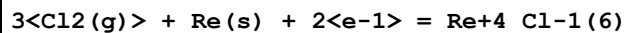
Reaction No. 67222							
$2\text{Re(s)} + 5\text{Te(s)} = \text{Re}_2\text{Te}_5\text{(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.91 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:12.81 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:8.696 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:6.206 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:4.519 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-17.61 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-22.50 (4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-22.66 (4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-22.96 (4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-23.39 (4SF)	4	MTD	6422

Reaction No. 71027							
$\text{ReO}_4\text{-1} + 4\text{H}_2\text{O} + 7\text{e-1} = \text{Re(s)} + 8\text{OH-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-68.1 (3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.584 (3SF)	0	CRV	6330

Reaction No. 72376							
$2\text{O}_2\text{(g)} + \text{Re(s)} + \text{e-1} = \text{ReO}_4\text{-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

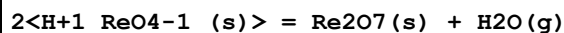
1	25	0	Inf. Dilution	lgK:122.3(4SF)	2	EOR	
2	25	0	Inf. Dilution	dG:-166.0(4SF)	0	CRV	6488
3	25	0	Inf. Dilution	dG:-694.5(4SF)kJ	3	CRV	7421
4	25	0	Inf. Dilution	dG:-166.000(6SF)	0	CRV	9029
5	25	0	Inf. Dilution	dG:-166.00(5SF)	0	CRV	10174
6	50	0	Inf. Dilution	dG:-167.20(5SF)	0	CRV	10174
7	75	0	Inf. Dilution	dG:-168.41(5SF)	0	CRV	10174
8	100	0	Inf. Dilution	dG:-169.62(5SF)	0	CRV	10174
9	125	0	Inf. Dilution	dG:-170.83(5SF)	0	CRV	10174
10	150	0	Inf. Dilution	dG:-172.03(5SF)	0	CRV	10174
11	175	0	Inf. Dilution	dG:-173.22(5SF)	0	CRV	10174
12	200	0	Inf. Dilution	dG:-174.40(5SF)	0	CRV	10174
13	225	0	Inf. Dilution	dG:-175.54(5SF)	0	CRV	10174
14	250	0	Inf. Dilution	dG:-176.64(5SF)	0	CRV	10174
15	300	0	Inf. Dilution	dG:-178.59(5SF)	0	CRV	10174
16	25	0	Inf. Dilution	dH:-188.2(4SF)	1	CRV	6488
17	25	0	Inf. Dilution	dH:-188.200(6SF)	1	CRV	9029

Reaction No. 79466



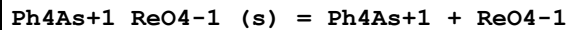
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-589.4(4SF)kJ	2	CRV	7421

Reaction No. 79672



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:4.460(0.018SD)	5	ETR	29294
2	25	0	Inf. Dilution	dH:11.020(0.100SD)	5	ETR	29294

Reaction No. 80966



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-8.49(3SF)	4	MSL	32204

Data Assessment

Weight	Assessment
9	highest accuracy
8	multi-determined; excellent dependability
7	trustworthy; outstanding single determination
6	better than average reliability
5	average single determination
4	some doubt
3	poorly known
2	guess to achieve consistency (or just as bad)
1	no information but consistent
0	problematic/inconsistent

Techniques

Abbrev	Method
CRV	Critical review
EAE	Automated extrapolation
EES	Personal estimate
EFE	Free Energy relations
ELC	Linear Combination of Reactions
ENS	Not specified
EOR	Other Reaction Constants
ETR	Thermodynamic relationships
MAG	Silver electrode e.m.f.
MCN	Conductivity
MDG	Combination of Thermodynamic Data
MEF	Electromotoric force, not specified
MIX	Ion exchange
MKN	Rate of reaction
MPS	Potentiometry and Spectrophotometry
MSL	Solubility
MSP	Spectroscopy
MTD	Temperature difference

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